# Supplementary Material

Technology Mentions and Direct Qualification Expressions in Embodied AI Recruitment

## Corpus construction and provenance

The initial wave required readable public job details, stable identifiers and core fields; it yielded eight firms and 610 advertisements. The expansion considered the remaining entities in the existing company landscape using organizational identifiability, research relevance, source quality and a current official board with a defensible boundary. Failed sources were recorded before the next feasible candidate was considered. The resulting 1,654 advertisements are a purposive snapshot, and neither wave provides an exhaustive ranking of excluded firms.

The 23-company comparison excludes the 24-record Amazon Frontier AI and Robotics team because it is bounded within Amazon. The 84-advertisement work-object case set covers machine-learning and artificial-intelligence (ML/AI) and software functions at four earlier case-priority firms; no preserved outcome-blind rule supports generalizing from those four. The 248-advertisement task sample instead applies a recorded function-by-primary-task intersection. Its technology catalog was fixed after an earlier pilot, and previous findings were known when the revision protocols were recorded. All analyses are exploratory.

Composite identity joins corpus, source advertisement ID and text-version hash. It produces exact matches for all task advertisements and avoids a duplicated convenience ID outside the task sample. Headquarters comes from company metadata, location from the captured advertisement or accepted reference, and language from source text. Six department-only location values are retained but excluded from geographic coverage. The language rule uses title plus the fullest body field, leaves strings with fewer than 40 Han-plus-Latin letters unclassified, and requires a unique leading Latin-language marker score. Country and location do not enter this classifier, and no advertisement was translated for coding.

## Functions activities and task membership

Primary function follows the continuing deliverable. Composite records contain several independent deliverables of similar weight; multiple-direction records explicitly recruit into alternative directions. Ambiguous, unmapped and source-insufficient cases remain visible. Activities are coded separately: research requires a concrete problem or object plus methodological exploration or scientific testing; engineering requires implementation, integration, debugging or operational delivery; and quality includes verification, compliance, release decisions or defect closure. Work objects in the four-firm case set may overlap, whereas primary task for the technology analysis is single-valued and follows the role's main responsibility.

The author reviewed the accepted corpus and settled disputed category designs and central rules, including attention to what the role actually delivers and does. The preserved release supports this project-level account but not a row-level independent second code. Historical model-review depth remains 469 full-text, 760 boundary-review and 347 carried-forward records for specific function, with 56/164/28 in the task sample. These strata do not become human-review counts.

## Qualification rules

Mention requires an explicit source occurrence. GENERAL and REQUIRED use both heading scope and local modifiers. Ordinary headings such as "Requirements", "Qualifications" or "Skills and Requirements" are

general; only an explicit required/must/mandatory scope makes an occurrence *R*. A statement under *R* is not automatically duplicated into *G*, although distinct statements can make both positive. Choice, example and preference operators propagate over their governed items, and a local operator overrides a broad heading. A separate unconditional occurrence remains positive.

The target is exact. C++ is not inferred merely from C; Linux is not inferred from Unix; Altium does not include unnamed printed-circuit-board (PCB) tools. The electromagnetic-compatibility, signal-integrity and power-integrity (EMC/SI/PI) indicator is the sole prespecified family and unions named compatibility, interference, immunity, signal-integrity and power-integrity methods once per advertisement. *U* is used for unresolved source scope, not as a generic response to reader disagreement. *Z* means no positive within adequate captured source and the rule, not employer absence. *Q* is derived from *G* and *R* only after each constituent is resolved.

## Diagnostic execution and adjudication

The diagnostic used seed 20260921 to order records within each task, technology and prior-*G* stratum. Up to five cases were retained from each of positive, unresolved, nonpositive-with-mention and no-mention strata, producing 105 pairs and 94 advertisements. This deliberately enriches boundary states; no probability weight is assigned to the match counts.

On 21 September 2026, GPT-5.6 Sol read the seven source packets under the saved rule sheet. Composite identity, employer metadata, earlier states, rationales and stratum were absent; target, task, title, original-language text and mechanically redacted source fields were visible. The reader also had prior exposure to a mandatory workspace summary containing aggregate project history. Its 105 judgments were frozen at SHA-256 `8f1ef32800186617347ae3af7e3109e48a0c5ac34ca8b1b5b23ca9e45dd25bbb` before comparison. The rule-sheet hash is `bb5948cdab7131a2614c073a9668b990ec9e9254f41436f4ed895f8f706784d2`; the consolidated packet hash is `83d8787cde2c092e20ba9464cb7c6dc3d97516f9231c3ea7aa24a6e6741bad1e`.

GPT-6 Astra, in the coordinating task, then reread all 16 *Q* differences and six additional *G/R* differences. The revision hierarchy is evidence based: three explicit-example PyTorch cases enter the current primary layer; ten slash/list cases retain the primary adjudication but admit the sole observed defensible alternative; three scope-inconsistent *Q* readings and six *Q*-unchanged heading/bucket readings enter only the all-fresh stress test. Five task flags cover four advertisements. Three advertisements flagged only for task support were not independently reread by the coordinating adjudicator; their flags are preserved for a membership restriction. Deployment labels are reported as observed. No backend snapshot, decoding parameter or model-generation seed is available.

The ten contested records form seven switches. C/C++ is coupled across A043/A080; the two targets in the same Python/PyTorch clause are coupled; and two advertisements using the identical EMC “including” template are coupled. Four singleton switches cover a trailing-等 skills list, Qt/C++, PyTorch/TensorFlow as choice, and Linux/Unix as a joint list. For the last, Linux must be directly named under the chosen slash reading: Unix is never substituted for Linux. Enumerating all $2^7=128$ coherent combinations gives a semantic-reading envelope without inventing a third state or attaching probabilities. This envelope is kept distinct from the *U*-assignment bounds within each combination. The all-fresh scenario substitutes all 105 new *G/R* outputs and is labelled an extreme stress test because it also imports readings rejected by the fixed rules. Exact quotes, reasons, coupling and hashes appear in the provenance ledger and process table.

## Weighting comparisons and local decomposition

Advertisement-weighted estimates describe observed advertisements; firm-equal estimates average valid firm shares. Eligibility requires minimum evaluable counts separately for *M*, *G* and *Q*, at thresholds ten and five. We report the corresponding *B*-only coverage so that exclusion due to unknown states is visible. Scenario comparisons use both dynamically eligible pairs and the fixed corrected-primary keys; identity changes cannot be mistaken for value changes.

The Galbot–Figure decomposition uses 82 Galbot Moka advertisements observed on 11 September and 60 Figure Greenhouse advertisements observed on 31 August. All contain captured body text and enter its activity denominator. Galbot sources are Chinese; Figure contains 51 English, four Spanish and five Portuguese advertisements. A symmetric Kitagawa identity decomposes the pooled difference into function composition and within-function rates. These checks concern the 142 local records and do not alter the whole-board source-quality screens, campus/internship exclusion or other previously defined sensitivities.

## Source diagnostic and adjudication layers

The source-reading diagnostic contains 105 advertisement–technology pairs from 94 advertisements. Relative to the frozen row states, the fresh reader matches 88 pairs for *G*, 100 for *R* and 89 for the union *Q*; 16 *Q* readings differ. These are state matches within the selected diagnostic records. They are not an accuracy estimate because the fresh model is not an independent gold standard and the unweighted diagnostic strata do not estimate a corpus error rate.

The 16 *Q* differences separate into three evidence layers. First, A032, A065 and A099 are clear example-scope cases and enter the corrected primary layer as *U*-to-*Z* changes for both *G* and *Q*. Second, ten list or slash cases remain contestable under the rule sheet: three retain their original list reading and seven retain unresolved slash scope. These ten pairs form seven coupled switches because two C/C++ readings, a Python/PyTorch stack in one advertisement and two identical EMC template readings must move together. Third, two fresh readings would disregard explicit example scope and one would disregard a local modifier; these remain outside the admissible set and enter only the all-fresh extreme. Five additional fresh *R* readings treat a generic *Requirements* heading as explicitly mandatory, and one fresh *G* reading includes a statement under *Required Skills*; none changes the corresponding *Q* state.

**Supplementary Table S1. Source diagnostic state matches**

| Task and technology | Sample pairs | G matches | R matches | Q matches | Q differences | Clear examples |
|---|---|---|---|---|---|---|
| Robot learning Python | 20 | 16 | 19 | 17 | 3 | 0 |
| Robot learning PyTorch | 20 | 15 | 20 | 15 | 5 | 3 |
| Robot runtime C++ | 20 | 17 | 19 | 17 | 3 | 0 |
| Robot runtime Linux | 15 | 13 | 15 | 13 | 2 | 0 |
| Board circuitry Altium | 15 | 14 | 12 | 14 | 1 | 0 |
| Board circuitry EMC/SI/PI | 15 | 13 | 15 | 13 | 2 | 0 |

Notes: The sample has 105 pairs in 94 advertisements. Counts compare frozen original labels with the frozen fresh output before adjudication. Prior-state stratification does not itself rule out probability sampling; these unweighted counts are not corpus accuracy estimates, and the fresh reader supplies no gold standard. The pooled matches are 88 for G, 100 for R and 89 for Q. The state matrices are in source_v2/analysis/diagnostic_confusion.csv; the full stratum frame is in source_v2/work/diagnostic_sampling_frame.csv. Source: calculations from the captured job-advertisement corpus.

## Examples of REQUIRED only and clarified expressions

Five advertisement–technology pairs in four advertisements are positive only under an explicitly required scope. The Chinese excerpts below are followed by labelled translations; the two English excerpts are reproduced verbatim.

- **E0206, PyTorch:** under “硬性要求” [mandatory requirements], “深度学习框架熟练(PyTorch 为主);能独立推进实验闭环。” [translation: “Proficient in deep-learning frameworks (primarily PyTorch); able to independently carry an experiment through a closed loop.”] PyTorch is the principal framework, so *R* and *Q* are positive while *G* is not.
- **E0205, PyTorch:** under “硬性要求” [mandatory requirements], “PyTorch 熟练;有仿真(Isaac / MuJoCo)或真机联调经验。” [translation: “Proficient in PyTorch; experience with simulation (Isaac/MuJoCo) or physical-robot integration.”] The choice after the semicolon does not govern PyTorch.
- **N0874, Python:** under *Required Skills*, “Strong Python programming skills.” This gives $R=Q=P$ and $G=Z$.
- **N0878, Python and PyTorch:** under *Required Skills*, “Strong Python skills and deep experience with PyTorch”. The conjunctive statement contributes two technology pairs in one advertisement. The required heading does not also make either pair a *G* positive.

The three primary clarifications concern explicit example cues. A032 and A099 contain “熟悉 PyTorch/TensorFlow 等深度学习框架” [translation: “Familiar with deep-learning frameworks such as PyTorch/TensorFlow.”]. A065 contains “有使用 PyTorch/TensorFlow、Isaac Gym/Mujoco 等工具进行机器人控制相关算法开发的经验。” [translation: “Experience using tools such as PyTorch/TensorFlow and Isaac Gym/MuJoCo to develop robot-control algorithms.”]. In each complete source, “等” governs an illustrative class and no separate unconditional PyTorch expression appears. The revision therefore resolves direct-expression status from *U* to *Z* without removing the mention.

A contrasting disputed source is N0149: “熟练使用 PyTorch/TensorFlow” [translation: “Proficient in using PyTorch/TensorFlow.”]. The slash has no explicit conjunction, choice or example marker. The primary layer retains unresolved scope; the alternative nonpositive reading is one of the admissible switches. This distinction explains why explicit examples are corrected while bare slash constructions remain a sensitivity question.

## Primary and sensitivity results

The immutable frozen version has *Q=170P/277Z/49U*. Applying the three clear example corrections produces the current primary states, *170P/280Z/46U*. The confirmed-positive share remains 170/496 (34.3 per cent), while the conditional proportion changes from 170/447 (38.0 per cent) to 170/450 (37.8 per cent). The share among mentions remains 170/348 (48.9 per cent); its upper assignment endpoint falls from 62.9 to 62.1 per cent.

The admissible sensitivity analysis applies every combination of the seven coupled switches to the ten contested pairs. All 128 readings preserve the same pair identities at both thresholds. The all-switch endpoint has *Q=179P/278Z/39U*; across the main comparisons, Python remains supported, PyTorch remains unresolved and Linux remains supported. At the threshold of five, 22 supported retentions and five supported reversals appear under every reading, while one additional reversal is supported under only some readings.

The all-fresh extreme has *180P/277Z/39U*. It additionally adopts three readings rejected by the fixed scope rules and the six fresh *G/R*-only differences, so it is an extreme stress test rather than another admissible adjudication. The task-boundary restriction begins from the corrected primary layer, excludes four advertisements represented by five diagnostic pairs, and leaves 244 advertisements and 488 pairs with *Q=169P/274Z/45U*. It does not alter either threshold's pair identities or four-category counts.

**Supplementary Table S2. State totals by evidence layer**

| Scenario | Ads | Pairs B | Q positive | Q nonpositive | Q unresolved | Q evaluable |
|---|---|---|---|---|---|---|
| Frozen original | 248 | 496 | 170 | 277 | 49 | 447 |
| Corrected primary | 248 | 496 | 170 | 280 | 46 | 450 |
| All admissible switches | 248 | 496 | 179 | 278 | 39 | 457 |
| All-fresh extreme | 248 | 496 | 180 | 277 | 39 | 457 |
| Task restriction | 244 | 488 | 169 | 274 | 45 | 443 |

Notes: The corrected primary applies only three explicit-example U-to-Z changes. The all-switch case is one of 128 admissible readings. The all-fresh extreme also adopts rejected scope and G/R readings and has a different evidential status. The task restriction starts from the corrected primary and removes four advertisements from both technologies; excluded rows remain identifiable in scenario_members.csv. Source: calculations from the captured job-advertisement corpus.

## Firm cells pair keys and offline navigation

`analysis/current_firm_cells.csv` contains all 120 current firm–task–technology cells: 44 for robot learning and model development, 40 for robot runtime and 36 for board circuitry. Each row supplies *B*, *M/G/R/Q* state counts, confirmed and unresolved shares, conditional proportions, assignment bounds and eligibility flags. All 120 cells have a fixed-base confirmed share. Three C++ cells have $Q_U=B$, so their conditional proportions are undefined and their bounds span 0–100 per cent.

`analysis/current_firm_pairs.csv` contains all eligible current comparisons. It has 35 threshold-specific rows: three at ten and 32 at five. The three main rows are included again at five; technologies reuse firms and

advertisements, so these rows are not independent observations. `analysis/main_company_comparisons.csv` isolates the three main comparisons, and `analysis/current_pair_summary.csv` supplies the six technology rows and total used in Table III. `analysis/current_task_summary.csv` supplies Table I and the within-mention composition for Figure 1. `analysis/ad_technology_current.csv` is the advertisement-level corrected layer; it preserves the historical state and revision identifier for the three changed rows.

Sensitivity files should be joined by the fixed pair key `threshold + task + label_code + company_a + company_b`. `analysis/admissible_reading_switches.csv` defines the 128 switch combinations. `analysis/admissible_reading_pairs.csv` and `analysis/admissible_reading_pair_summary.csv` contain their pair results; `analysis/admissible_comparison_envelopes.csv` gives cross-reading envelopes for each fixed comparison; and `analysis/admissible_classification_ranges.csv` gives the reported category ranges. The `analysis/scenario_*` files contain the frozen, corrected-primary, all-switch, all-fresh and task-restriction scenarios. `analysis/scenario_key_audit.csv` verifies that no scenario adds or removes a threshold-five or threshold-ten pair key.

An independent recomputation in `verification/independent_pairs.csv` and `verification/independent_analysis.json` reads only the two frozen v2 inputs, applies A032/A065/A099 directly, rebuilds all 120 cells and enumerates the company pairs. It reproduces the 3/32 pair counts, current category totals and main difference bounds. This is an arithmetic and transformation check; it does not independently validate source meaning.

**Supplementary Table S3. Joint comparison classifications across named scenarios**

| Scenario | Min N | Pairs | Supported retention | Supported reversal | Conditional reversal with overlap | Same direction with overlap |
|---|---|---|---|---|---|---|
| Frozen original | 5 | 32 | 22 | 5 | 3 | 2 |
| Frozen original | 10 | 3 | 2 | 0 | 1 | 0 |
| Corrected primary | 5 | 32 | 22 | 5 | 3 | 2 |
| Corrected primary | 10 | 3 | 2 | 0 | 1 | 0 |
| All admissible switches | 5 | 32 | 22 | 6 | 1 | 3 |
| All admissible switches | 10 | 3 | 2 | 0 | 0 | 1 |
| All-fresh extreme | 5 | 32 | 22 | 6 | 1 | 3 |
| All-fresh extreme | 10 | 3 | 2 | 0 | 0 | 1 |
| Task restriction | 5 | 32 | 22 | 5 | 3 | 2 |
| Task restriction | 10 | 3 | 2 | 0 | 1 | 0 |

Notes: Each scenario independently reapplies eligibility and retains exactly the corrected-primary pair identities: three at ten and 32 at five. Across all 128 admissible readings, 22 supported retentions and five supported reversals persist at five; a sixth supported reversal occurs in some readings. The main Python and Linux directions remain supported and PyTorch remains unresolved. These are deterministic sensitivity results, with no probability weights. Source: calculations from the captured job-advertisement corpus.

## Engineering activity decomposition as background

The earlier Galbot–Figure engineering-activity decomposition remains a supplementary composition example. Its fixed four-function comparison contains 82 Galbot and 60 Figure advertisements. Galbot's pooled engineering-activity rate exceeds Figure's by 11.10 percentage points, decomposed arithmetically as *+19.45-8.35*: a positive function-composition component and a negative within-function component. The identity describes these advertisements and does not estimate a firm effect.

The local source audit and the whole-board R1/R2/R3 screens answer different questions. All 142 records in the four-function comparison have captured text and evaluable activity status, establishing denominator recoverability rather than semantic accuracy. R1, R2 and R3 instead retain 17 firms/968 advertisements, 11/576 and 11/557 after successively applying whole-board activity evaluability, source-quality and description-length rules. Galbot fails those broader board screens even though its 82 records in the local example are evaluable.

## Firm coverage tables

**Supplementary Table S4. Task coverage and pooled and firm-equal summaries**

| Task and technology | Firms with task | Firms with evaluable Q | Pooled confirmed % | Firm-equal confirmed % | Pooled conditional % | Firm-equal conditional % |
|---|---|---|---|---|---|---|
| Robot learning Python | 22 | 22 | 40.0 | 40.0 | 47.7 | 44.7 |
| Robot learning PyTorch | 22 | 22 | 21.9 | 21.3 | 23.0 | 22.1 |
| Robot runtime C++ | 20 | 17 | 43.7 | 34.4 | 59.4 | 51.5 |
| Robot runtime Linux | 20 | 20 | 54.0 | 45.7 | 54.7 | 46.0 |
| Board circuitry Altium | 18 | 18 | 8.9 | 8.8 | 8.9 | 8.8 |
| Board circuitry EMC/SI/PI | 18 | 18 | 26.8 | 28.2 | 26.8 | 28.2 |

Notes: Confirmed shares use B and include all firms with task advertisements. Conditional proportions use B−U and exclude only firms with zero evaluable Q; zero-positive firms otherwise remain. There are 20 runtime firms but only 17 evaluable C++ firms. Firm-equal averages weight eligible firms equally; pooled summaries weight advertisements. Different denominators must not be treated as interchangeable. Source: calculations from the captured job-advertisement corpus.

**Supplementary Table S5. Eligibility loss relative to the full task base**

| Minimum N | Cells with B ≥ N | Cells passing M G Q | Excluded by evaluability | Additional G exclusions | Eligible pairs |
|---|---|---|---|---|---|
| 5 | 26 | 21 | 5 | 0 | 32 |
| 10 | 10 | 9 | 1 | 0 | 3 |

Notes: The electronic companion tables/S5_Full_Firm_Coverage.csv reports all 120 cells, including B, U/B, M_N, G_N, Q_N and both eligibility flags. Retaining G maintains a common comparison frame; it excludes no additional cell beyond M and Q in these data. A cell may pass the threshold without another eligible firm in its task–technology combination, so eligible cells do not directly equal pair counts. Source: calculations from the captured job-advertisement corpus.

## Diagnostic process tables

**Supplementary Table S6. Actors and timing of the source diagnostic**

| Stage and timing | Actor or mechanism | Material and evidential limit |
|---|---|---|
| Corpus review Retrospective statement 21 September 2026 | Author | Full accepted corpus reviewed; composite and alternative-direction categories, later categories and deliverable-based rules confirmed. No independent second technology code is documented for every pair. |

| Stage and timing | Actor or mechanism | Material and evidential limit |
|---|---|---|
| Sample and packet<br>21 September 2026 | Coordinating task; deterministic hash selection | Up to five records in four prior-G strata per task–technology cell. Sampling seed 20260921 is not a model-generation seed. |
| Fresh source reading<br>21 September 2026 | GPT-5.6 Sol deployment | 105 target pairs, 94 advertisements, rule sheet and source fields. Row labels, rationales, composite identities, firm metadata and strata withheld. High-level workspace history was visible. Output frozen before comparison. |
| Source adjudication<br>21 September 2026 | GPT-6 Astra deployment in coordinating task | All 16 Q differences and six further G/R differences reread against captured source. No independent human gold standard. Three task-only flagged advertisements did not receive a documented additional full rereading. |
| Revision and sensitivity<br>22 September 2026 | Coordinating analysis following the revision request | Three explicit-example corrections; seven coupled switches for ten disputed Q readings; rejected fresh readings isolated in an extreme test. No new source-reading sample was generated. |
| Reproducibility record | Saved task and file provenance | Exact neutral prompt, packets, fresh outputs, row excerpts, decision ledger and SHA-256 hashes accompany the package. Deployment labels are recorded; backend snapshots and decoding settings were not preserved. |

Notes: The detailed nine-stage process and complete model prompt provenance are in work/provenance/diagnostic_process.md. The author's retrospective project review is distinct from model source reading and arithmetic verification. Source: calculations from the captured job-advertisement corpus.

**Supplementary Table S7. Coupled admissible source readings**

| Pair identifiers | Primary to alternative Q | Alternative reading |
|---|---|---|
| A043, A080 | A043: U → P<br>A080: U → P | C/C++ jointly names C and C++ in both constructions; C++ remains the target. |
| A046 | A046: Z → P | The trailing skills list is an inclusive enumeration of directly specified skills. |
| A049, A050 | A049: U → P<br>A050: U → P | Python/PyTorch jointly names an implementation stack. The two technology readings within the same advertisement change together. |
| A067, A069 | A067: Z → P<br>A069: Z → P | The repeated including-list qualification directly includes the named method in both advertisements. |
| A083 | A083: U → P | Qt/C++ jointly names an implementation stack. The separate task-membership question enters only the task-restriction scenario. |
| A097 | A097: U → Z | The bare PyTorch/TensorFlow pair offers a choice between frameworks. This is the sole observed alternative state used. |
| A104 | A104: U → P | Linux/Unix is a joint or inclusive list directly naming Linux. This interpretation does not substitute Unix for the fixed Linux target. |

Notes: Each row is one binary switch, yielding 128 combinations. Repeated constructions and two targets in one advertisement switch together as indicated. The fixed target remains unchanged in every scenario. The exact original-language quotes and the three primary corrections, ten disputed Q readings, three rejected Q readings and six further G/R differences are recorded in work/provenance/diagnostic_adjudication_ledger.csv. Source: calculations from the captured job-advertisement corpus.